



cobas e 411 analyzer

Host Interface Manual

Version 2.1

Revision history

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Changes V 2.1 additional screenshot and remark on page 11
Host teste code setting

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Overview

Special Note

This document describes communication procedure related to communication method that enables intercommunication between Laboratory Host System, hereinafter referred to as HOST, and **cobas e 411** analyzer. Specification and software described herein comply with the following ASTM communication protocol (HOST communication ASTM higher-layer: High-Level, lower-layer: Low-Level I/F specification): Specification X12 of ASTM (American Society of Testing and Materials)

ASTM E1381-91: Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer System

ASTM E1394-91: Standard Specification for Transferring Information between Clinical Instruments and Computer System

As **cobas e 411** analyzer is a succeeding version of the **Elecsys® 2010** analyzer, it upholds **Elecsys® 2010** analyzer HOST communication protocol. Further, a new communication protocol is added to keep compatibility with **cobas®**. These two (2) communication protocols such as "**Elecsys®** type" and "**cobas®** type" are selectable. MSRs (Manufacturer Specific Records) that are the original protocol of **Elecsys® 2010** analyzer are not supported by **cobas e 411** analyzer.

In case of communication problems between a host and the **cobas e 411** analyzer please copy the trace file d:\e411\data\log\host_trace.log onto a removable media and send it to technical support.

Specification of interface

B

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Specification of interface

Communication between **cobas e 411** analyzer and HOST is based on RS-232C connection.

Specification of communication

Specification of serial interface is shown as follows:

Table B-1 Specification of serial interface

Item	Specification	Recommendation	Note
Communication Protocol	ASTM	-	
Communication Speed	19200bps/9600bps/4800bps	9600bps	Selectable from GUI
Character Configurations	See Table 2-2.	8bit,NONE,1Stopbit	Selectable from GUI
Protocol Type	Elecsys® type or cobas® type	Elecsys® type	Selectable from GUI
Frame Length	247 bytes	-	
Communication Port	1 port	-	
Cable Length (maximun)	15m	-	
Communication method	Half duplex		

Character configuration is selected from Table 2-2.

Table B-2 Character configuration

No.	Data bit	Parity bit	Stop bit
1	7 bit	EVEN	2 stop bit
2	7 bit	ODD	2 stop bit
3	7 bit	EVEN	1 stop bit
4	7 bit	ODD	1 stop bit
5	8 bit	NONE	2 stop bit
6	8 bit	NONE	1 stop bit
7	8 bit	EVEN	1 stop bit
8	8 bit	ODD	1 stop bit

Host communication setting

The screenshot displays the 'Start Conditions' window of the Roche/Hitachi cobas e 411 analyzer. The interface includes a top status bar with 'Host E Stand-By', 'bm serv', the date '2007/08/05 (Mon.)', and the time '15:46'. The main window has a blue header 'Start Conditions'. Inside, there are three main sections: 'Disk No.' with a value of '9', 'Host Setting' with 'Host Communication' set to 'On' (with 'Off' also available), and 'Sample Reception Mode Setting' with '99 Hour' and 'On' (with 'Off' also available). A large blue 'START' button is on the right. At the bottom left are 'Cancel' and 'OK' buttons. The bottom status bar contains a 'Help' button, 'Guidance' text, and a 'Print Screen' button. A vertical column of buttons on the right side includes 'Stop', 'Logoff', 'S.Stop', 'Stat Mode', 'Alarm', 'Print', and 'Start'.

Select "On" or "Off" at "Host Communication" on Start Conditions screen.

Select condition of HOST communication on [Utility]-[System]-[Host Communication Setting] window. The setting is changeable when "Host Communication" is off.

Host **E** Stand-by bmserv 09/22/2008 (Mon) 10:09

Workplace	Reagent	Calibration	QC	Utility
System	Maintenance	Application	Calc. Test	

Stop

Logoff

S.Stop

Stat Mode

Alarm

Print

Start

Host Communication Setting

Protocol Type: **Elecsys** **Cobas**

RS232C Setting: Speed **19200** Frame **7Bit,EVEN,2StopBit**

Automatic Result Upload: **Yes** **No**

Communication Trace: **Yes** **No**

Cobas type Setting:

Automatic Recovery of Session: **Yes** **No**

QC Additional Information: Control Name **Lot No.** **Host Test Code**

Send Results Only Mode: **Yes** **No**

Cancel OK

? Help Print Screen

Host **E** Stand-by bmserv 09/22/2008 (Mon) 10:15

Workplace	Reagent	Calibration	QC	Utility
System	Maintenance	Application	Calc. Test	

Stop

Logoff

S.Stop

Stat Mode

Alarm

Print

Start

Host Communication Setting

Protocol Type: **Elecsys** **Cobas**

RS232C Setting: Speed **19200** Frame **7Bit,EVEN,2StopBit**

Automatic Result Upload: **Yes** **No**

Communication Trace: **Yes** **No**

Cobas type Setting:

Automatic Recovery of Session: **Yes** **No**

QC Additional Information: Control Name **Lot No.** **Host Test Code**

Send Results Only Mode: **Yes** **No**

Cancel OK

? Help Print Screen

Host Test Code Setting

Test	Test Code	Host Code
TSH 0	10	
FT3 1	61	
CA15-3 2	332	
A-HBC 0	450	
B12 0	600	

Host Code

Update

Cancel OK

In cobas mode it is mandatory to enter also the Host Code to upload results.

Host Test Code Setting

Test	Test Code	Host Code
TSH 0	10	10
T4 1	21	20
FT4 0	30	30
TNTSTAT 1	201	200

Host Code

Update

Cancel OK

Transfer parameters are shown in Table B-3.

Table B-3 Transfer parameters list

Parameter	Option	
Protocol type	Elecsys® / cobas®	When selecting [Elecsys®], it communicates with HOST by Elecsys® type. When selecting [cobas®], it communicates with HOST by cobas® type.
RS232C setting		
Speed	19200/9600/4800	Select speed.
Frame	7 bit, EVEN, 2 stop bit/ 7 bit, ODD, 2 stop bit/ 7 bit, EVEN, 1 stop bit/ 7 bit, ODD, 1 stop bit/ 8 bit, NONE, 2 stop bit/ 8 bit, NONE, 1 stop bit/ 8 bit, EVEN, 1 stop bit/ 8 bit, ODD, 1 stop bit	Select frame.
Automatic result upload	Yes/No	When [YES] is selected, result data of a sample is sent to HOST in real time as soon as all such data are collected.
Communication trace	Yes/No	When [YES] is selected, communication detail with HOST is traced in cobas e 411 analyzer.
cobas® type setting		When Protocol Type is [cobas®], parameters are as follows.
Automatic recovery of session	Yes/No	This is an automatic reconnection function in case of communication is disconnected, (HOST Communication Off) under error described ASTM1381-91 has occurred. When [Yes] is selected, the erroneous message and a message sent from HOST before the reconnection process completed are rejected.
QC additional information	Control name/Lot No.	When [ControlName] is selected, name of control is sent as sample ID. When [LotNo.] is selected the Control Name and the Lot number is sent to the host, lot No. is sent as sample ID.
Send result only mode	Yes/No	Only result data is sent. Inquiry for order is not conducted. This function is not applied to Batch transfer.
Host test code	Host test code setting	Test code between cobas e 411 analyzer and Host defined in Host Test Code Setting in Utility/Host Communication Setting/Host Test Code. All used test codes must be entered, Hint: Also new test application has to be entered.

Communication cable

Connect RS-232C communication cable to the connector at the left side of **cobas e 411** analyzer. The connector is D-SUB9. Fig. B-1 shows connector and communication cable. Fig. B-2 shows connection diagram.



Fig. B-1 Connector

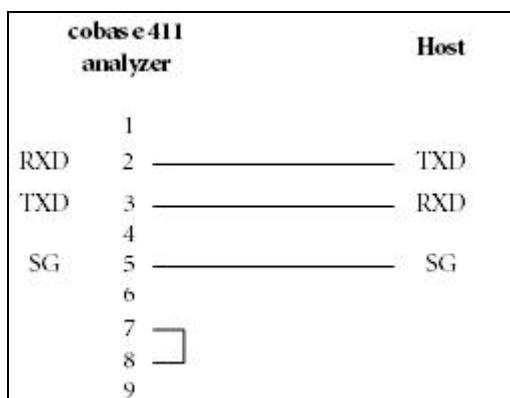


Fig. B-2 Connection diagram

1	<i>Communication text</i>	C-2
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Communication text

Table C-1 shows the communication text of the **cobas e 411** analyzer.

Table C-1 List of communication text

Text	Direction	Elecsys® type		cobas® type	
		Real time	Batch	Real time	Batch
Inquiry for order	Upload e 411 > HOST	Yes	No	Yes	No
Order for test request	Download HOST > e 411	Yes	Yes	Yes	Yes
Result report	Upload e 411 > HOST	Yes	Yes	Yes	Yes
Inquiry of result	Download HOST > e 411	No	No	No	No

(Yes: equipped, No: not equipped)

Note:

Patient sample and quality control sample are sent as result data, but not calibration result.

Note:

cobas e 411 analyzer is not equipped with auto-rerun function. Rerun sample can be manually defined.

Note:

There is no Inquiry for order of a control sample

Table C-2 shows Causes of communication text.

Table C-2 Causes of communication text

Text	Real/Batch	Cause
Inquiry for the requested tests (upload)	Real	◆ After sample ID is read, inquire of HOST for test selection information of patient sample to which test selection information is not registered. ◆ Wait replay from HOST for test selection information for a certain length of time after the inquiry. If not replied even after a certain length of time, cancel the inquiry.
Order for test request (download)	Real	◆ Specify the test selection information for a sample when the test selection information is inquired.
	Batch	◆ HOST specifies the test selection information of a patient sample at a given timing. Register test selection information before reading sample ID to use this function.
Result report (upload)	Real	◆ Send result data of patient sample and quality control sample when all test data of each sample is collected.
	Batch	◆ Send result data of patient sample and quality control sample specified on [DataReview] window.

ASTM communication protocol

D

1	<i>Communication data structure</i>	D- 2
2	<i>Frame structure</i>	D- 3
3	<i>Definition of communication protocol</i>	D- 5
4	<i>Flow chart</i>	D-10

ASTM communication protocol

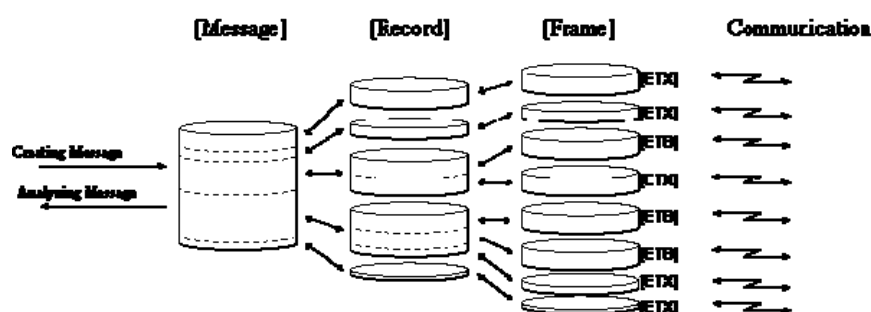
cobas e 411 analyzer employs ASTM communication protocol.

Communication data structure

ASTM communication protocol consists of three layered data structure such as message, record, and frame. Data is communicated by message. Further, data is communicated by frame actually. Data structure of a frame varies by protocol.

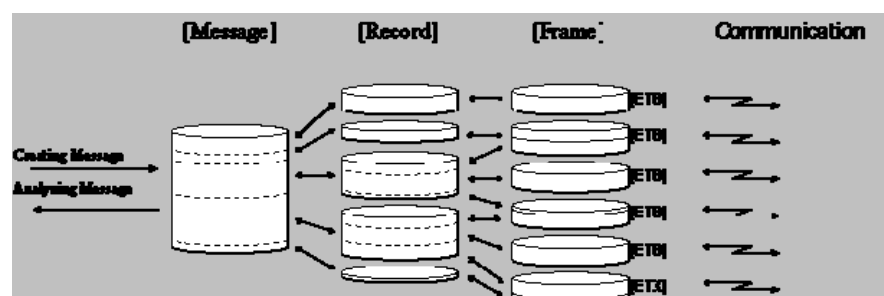
(1) Elecsys® type

A message consists of multiple records. A record consists of one or more frames. A frame comprises not more than one record. In case a record exceeds 240 bytes, a frame is divided into middle frames and a last frame. [ETB] is used for the middle frame and [ETX] is used for the last frame.



(2) cobas® type

A message consists of several records. A record consists of one or more frames. A frame may comprise multiple records. In case a record exceeds 240 bytes, a frame is divided into middle frames and a last frame. [ETB] is used for the middle frame and [ETX] is used for the last frame.



Frame structure

Frame structure is shown below.

- ◆ Middle frame, when a message is divided into more than one frame.

[STX]	FN	text	[ETB]	CS1	CS2	[CR]	[LF]
-------	----	------	-------	-----	-----	------	------

- ◆ When the last frame in a message or a frame is single frame.

[STX]	FN	text	[ETX]	CS1	CS2	[CR]	[LF]
-------	----	------	-------	-----	-----	------	------

Field	ASCII Code	Content	Character	Note
[STX]	0x02	Start of text	1byte	
FN	-	Frame No.	1byte	1
text	-	Communication data	Max. 240 byte	2
[ETX]/[ETB]	0x03/0x17	End of text/end of communication block	1byte	3
CS1	-	Check sum	1byte	4
CS2	-		1byte	
[CR]	0x0d	Carriage return	1byte	
[LF]	0x0a	Line feed	1byte	

*Note 1:

Way to assign frame No. (FN) is starting from No. 1 to No. 7. When exceeding No. 7, start from No. 0 to No. 7.

*Note 2:

Codes except the following ASCII code are available for text.

Code	Code	Code	Code	Code
[SOH]0x01	[STX]0x02	[ETX]0x03	[EOT]0x04	[ENQ]0x05
[ACK]0x06	[LF]0x0A	[DLE]0x10	[DC1]0x11	[DC2]0x12
[DC3]0x13	[DC4]0x14	[NAK]0x15	[SYN]0x16	[ETB]0x17

*Note 3: When a message is 240 bytes or less, use [ETX]. When exceeding 240 bytes, use [ETB].

*Note 4: Add each character code between and inclusive frame No.(FN) and [ETB] or [ETX].

Display the sum in hexadecimal format. Convert the last two digits into ASCII code. Code used for Check Sum is "0" to "9" and "A" to "F".

Ex. Check sum calculation method

[STX]	"1"	"T"	"e"	"S"	"t"	[ETX]
-------	-----	-----	-----	-----	-----	-------

Field	Character	Hex. format	Sum
[STX]	[STX]	02h	-
FN	"1"	31h	31h
text	"T"	54h	85h
	"e"	65h	EAh
	"s"	73h	15Dh
	"t"	74h	1D1h
[ETX]	[ETX]	03h	1D4h
			Last two digits of the sum. D4h
CS1	"D"	44h	
CS2	"4"	34h	
[CR]		0Dh	
[LF]		0Ah	

[STX]	"1"	"T"	"e"	"S"	"t"	[ETX]	"D"	"4"	[CR]	[LF]
-------	-----	-----	-----	-----	-----	-------	-----	-----	------	------

Definition of communication protocol

Low-Level Protocol of ASTM communication protocol is one-way. Response is generated after the information is sent. Response is not generated simultaneously with communication. Unlike the other communication protocols, it does not have master-slave relation. Both **cobas e 411** analyzer and HOST enable to initialize the communication. When establishing send system and receive system, or when having the action of both the sender and the receiver arranged properly, the information is communicated by the following three phases.

- ◆ Establishment Phase
- ◆ Transfer Phase
- ◆ Termination Phase

Establishment Phase

In data link layer, both sender and receiver go into one of the following status.

- ◆ Idle state: status waiting for becoming receiver.
- ◆ Starting Establishment Phase at the transmitting side (sender), by sending [ENQ].
- ◆ After receiving {ENQ} from the sender the receiver sends back [ACK].

Number of sender or receiver is one at a time. Without ongoing communication the **cobas e 411** and the HOST are waiting in idle state. When a message is sent from one side and the presentation layer requires the data link layer to send record, one side changes its status from idle to sender. To determine the direction of communication the sender starts the Establishment Phase by sending ASCII Code 05h [ENQ] and turns into status sender. The receiver answers this request by sending ASCII Code 06h [ACK] and turns into status receiver. Transfer Phase starts when Establishment Phase is completed by receiving [ACK]. Reception other than [ENQ] is ignored in idle status.

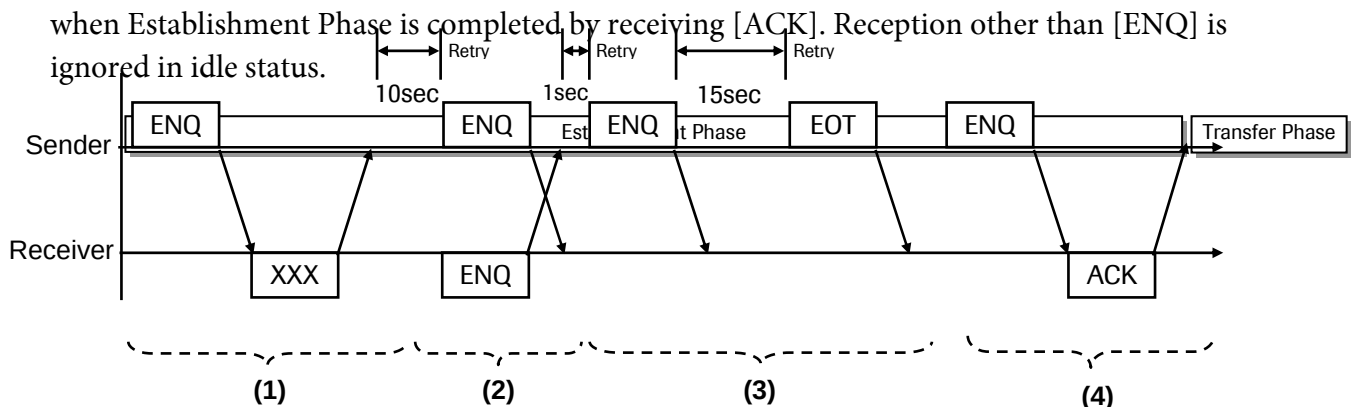


Fig D-1 Establishment Phase

In case of error, there are additionally three options for the receiver to respond [ENQ].

(1) Receiver sends characters other than [ACK].

These characters are normally sent by using ASCII code 15hex [NAK] when the receiver is busy. The sender waits for a certain length of time, e.g. **cobas e 411** analyzer waits for 10 seconds, and tries to establish with another [ENQ]. The **cobas e 411** analyzer repeats this cycle until the number of retries after error reaches six.

(2) Receiver sends [ENQ].

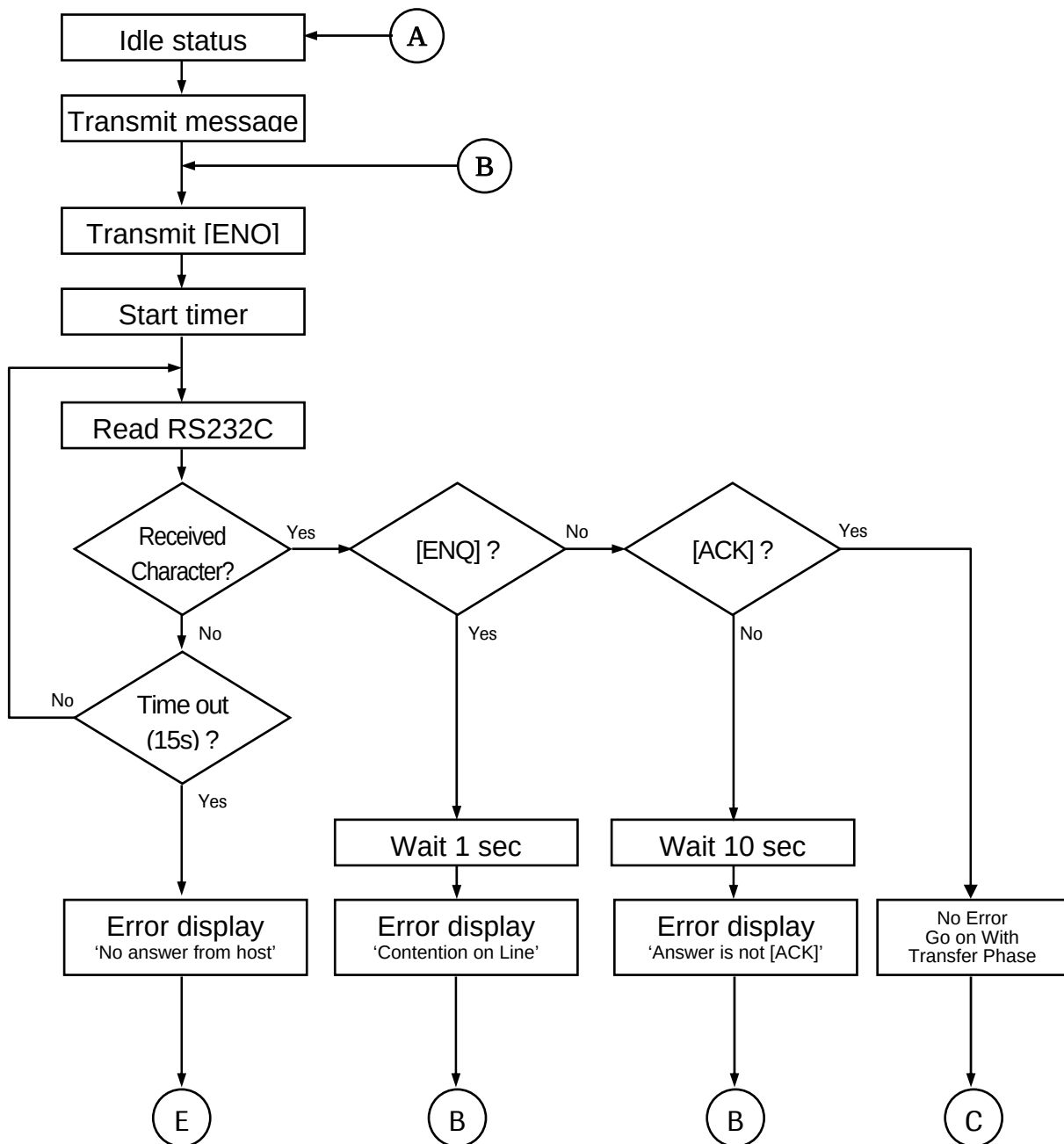
The status in which both sender and receiver are trying to change their status to the sender is called "Link Contention" in ASTM. When in link contention, it is defined that communication information of **cobas e 411** analyzer has a priority. So that HOST has to stop sending [ENQ] and has to respond simultaneously by [ACK] or [NAK] when the Link Contention is detected. On the other hand, the **cobas e 411** analyzer waits for 1 second and replies [ENQ]. The sender repeats this cycle until receiving characters such as [ACK] or [NAK].

(3) No response from receiver.

The sender starts Termination Phase by sending ASCII code 04hex [EOT] after waiting for 15 seconds, and displays an error message.

(4) Successful procedure of establishment Phase is mentioned above.

Establishment Phase



Transfer Phase After receiving frame, sender discontinues communication until receiving the response or occurrence of time-out. Usually the receiver notifies by sending [ACK] that it successfully received the last frame and completes its preparations to receive the next frame. The receiver notifies by sending [NAK] that the last frame was not received and it is waiting for receiving the frame.

HOST in status receiver

According to the above, there are three options for HOST to respond the communication of the frame.

◆ HOST sends [ACK]

cobas e 411 analyzer sends the next frame. If the **cobas e 411** analyzer has data that is to be communicated, **cobas e 411** analyzer continues to send frames.

◆ HOST sends characters other than [ACK]

cobas e 411 analyzer repeatedly sends the frame. This cycle is repeated until "number of retry in case of error" reaches six. At this moment, **cobas e 411** analyzer starts Termination Phase and displays an error message by sending [EOT].

◆ No response from HOST (timeout)

cobas e 411 analyzer starts Termination Phase by sending [EOT] after 15 seconds and displays an error message.

Response depends on how HOST responds to frame communication from **cobas e 411** analyzer.

HOST in status sender

When **cobas e 411** analyzer is in the receiving status and is waiting for communication from HOST, there are the following three scenarios.

◆ HOST sends frame characters.

After the complete frame is received, the frame No. and the checksum are checked if they are correct. When the frame is correct, **cobas e 411** analyzer responds by [ACK]. When the frame is incorrect, the incorrect frame is rejected and [NAK] is returned.

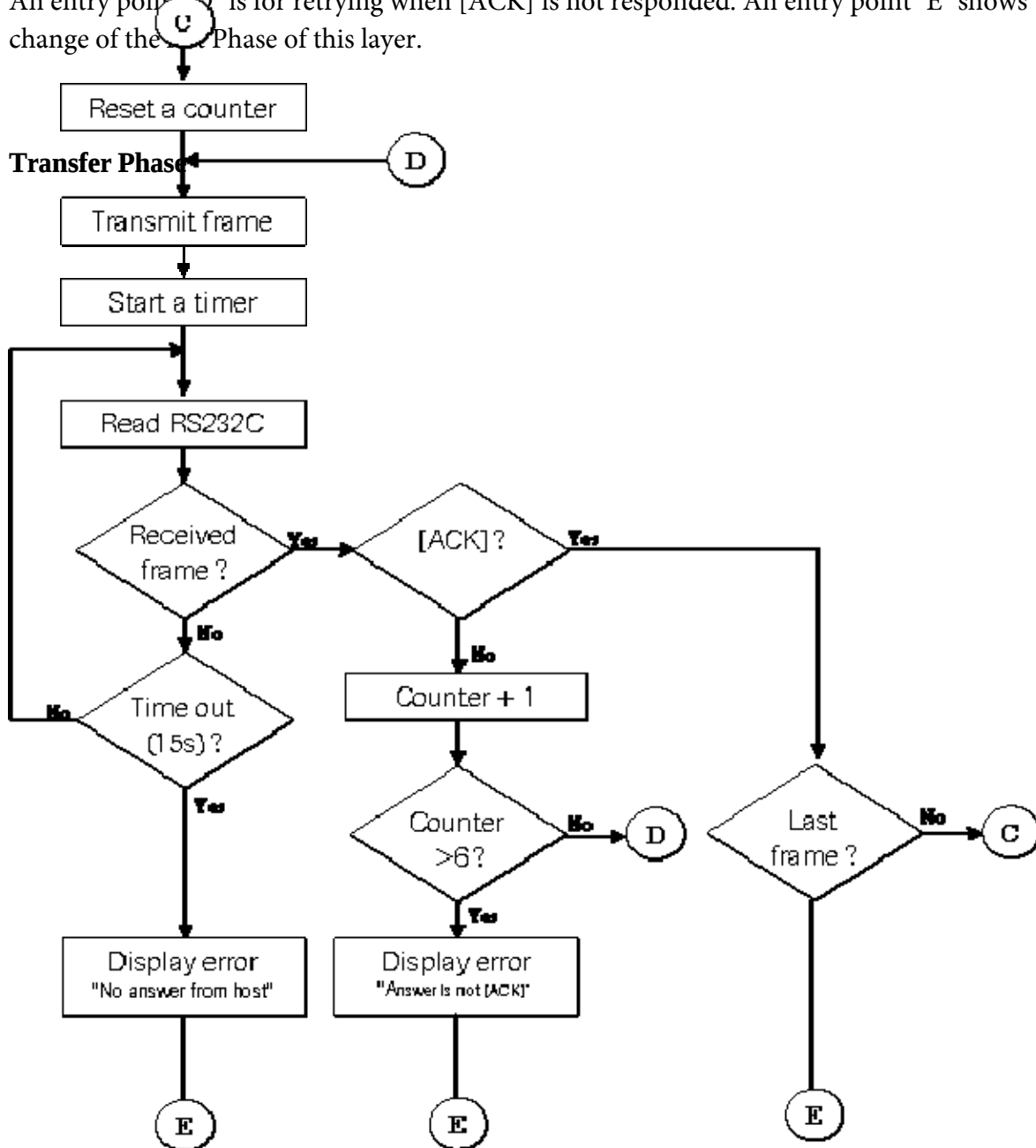
◆ HOST does not complete frame communication.

Time-out occurs when receiving unfinished frame and at the same time [EOT] is not received after 15 seconds counted from the last communication of [ACK] or [NAK] from **cobas e 411** analyzer. **cobas e 411** analyzer is turned into idle status by deleting the last incomplete message. The line is deemed to be neutral status.

◆ HOST sends [EOT].

cobas e 411 analyzer is turned into idle status. Only completely received frame is deemed to be effective.

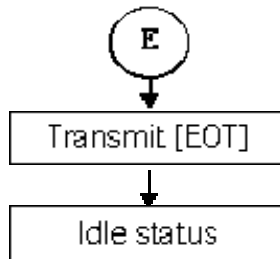
"Transfer Phase" shows the flowchart of three types of response against frame communication. An entry point "C" is selected when Establishment Phase is completed without any trouble. The entry point "C" is also the reentry point when the following frame communication is succeeded. An entry point "D" is for retrying when [ACK] is not responded. An entry point "E" shows change of the Transfer Phase of this layer.



Termination Phase

Both sender and receiver change their status into idle in Termination Phase. This Phase only starts when the sender sends [EOT]. Response from the receiver to this message is none. When [EOT] is detected at the receiver, it is turned to be idle and the line is required to be changed to neutral.

Termination Phase



1	<i>Syntax</i>	<i>E- 2</i>
2	<i>Message header record</i>	<i>E- 6</i>
3	<i>Message termination record</i>	<i>E- 7</i>
4	<i>Request information record</i>	<i>E- 8</i>
5	<i>Patient information record.....</i>	<i>E- 9</i>
6	<i>Test order record</i>	<i>E-10</i>
7	<i>Result record.....</i>	<i>E-13</i>
8	<i>Comment record.....</i>	<i>E-16</i>

Record structure

The cobas e 411 host protocol is compatible with the cobas 6000 protocol with some minor differences.

No.	Object	Remarks
1	Carrier no.	Cobas e 411 rack: When rack no. is unknown, “@” is attached in front of <CarrierNo> followed by a delivered value. Cobas e 411 disk: The carrier no. is always available on a disk system.
2	Sample Id	When barcode read error occurs, the sample id will be generated as follows: “@”<Sequence no.> If a sample barcode read error occurs on a rack system and the rack no. has been generated, the sample can not be identified by the host.
3	Conatiner Type	“MC” is not micro cup, but indicates reduced value.
4	Result Record	cobas® type format transmits calculated tests. The host test code numbers can be defined.
5	Comment Record (following the order record)	The comment record, that follows the order record for patient demographic data is not used.
6	Comment Record (following the result record)	When no data alarm for message value is available, no comment record will be transmitted to host.

Syntax

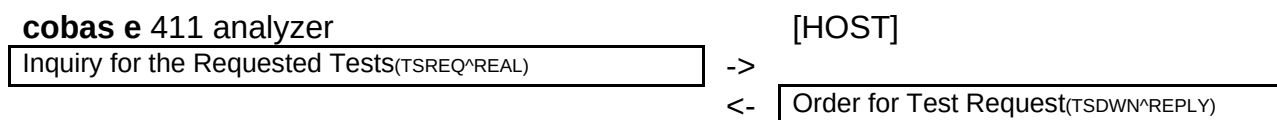
Syntax used in message (records) communicated by **cobas e 411** analyzer is shown below.

Table E-1 Message syntax

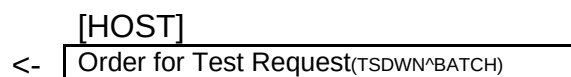
Communication text	Real/Batch	Message syntax	
		Elecsys® type	cobas® type Comment or Special Instructions
Inquiry for the requested tests (upload)	Real	H Q L	H Q L TSREQ^REAL
Order for test request (download)	Real	H P O L	H P O L TSDWN^REPLY
	Batch		L TSDWN^BATCH
Result report (upload)	Real	H P O L	H P O L RSUPL^REAL
	Batch	{ R C-RES }	{ R C-RES } RSUPL^BATCH

Communication sequences (message flow) of communication message by application layer are shown below.

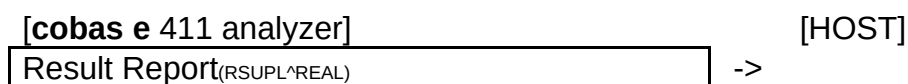
(1) Order for test request : real-time



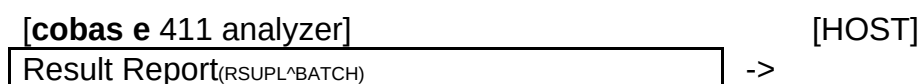
(2) Order for test request : batch



(3) Result report : real-time



(4) Result report : batch



ASTM syntax is shown below.

(a) Terminating and separating records character = "CR":

Indicates completion of record. ASCII character [CR] (0Dhex) is required.

(b) Field separator character = vertical bar "|":

Separates adjacent fields in a record. Also separates the first record ID.

Depending on the second character of a message title record, it enables to define the field separator character randomly. However "|" is recommended.

(c) Repetition field separator character = backslash "\":

When fields consist of repetition of the same data, it is called "repeated field." Repeated field separator character is a separator between tests of repeated field. Depending on the message title code, it enables to define the repeated field separator character randomly. However "\" is recommended.

(d) Component separator character = caret "^":

When fields consist of multiple components, it is called "component field." The component separator separates the components in a field. Depending on the message title code, it enables to define the component separator character randomly. However "^" is recommended.

(e) Escape character = ampersand "&":

Escape character is defined to represent a separator character in the field including normal text. Appearance of this character in such field indicates that the following character has special meaning. Depending on the message title code, it enables to define the escape character randomly. However "&" is recommended.

(f) Expression of special characters by escape characters:

Escape sequence, character string starts and ends with &, is defined as follows. When these sequences are detected in the field, it is interpreted the corresponding character string.

&F& Field separator character
&S& Component separator character
&R& Repetition separator character
&E& Escape character

Escape sequences other than the above are skipped and handled as null value.

The table below describes attributes of each field in each record shown in chapter 5.

Column	Name of Attributes	Description
1	Field position (Pos)	Field position. The number also increases by 1, if the field is not used.
2	Order (No.)	Sequence of the field. Sequence of the fields in record.
3	Name of Field (Field)	Name of relevant field.
4	Type (Type)	Typing characters for the fields are any of the following.
		ST Character string.
		TX Character string group that end is printable.
		NM Numeric value. "+" or "-" is attached at the top. If not, the value is deemed to be "+." When without decimal point, the value is deemed to be integer. Prefix attached to "0," and suffix attached to "0" of numeric value with decimal point can be anything.
		DT Date. Four digits of dominical year. YYYYMMDD (YYYY: dominical year, MM: month, DD: day) Ex. September 5, 1995 is displayed as "19950905."
		TM Set time in 24 hours. HHMMSS (HH: hour, MM: minute, SS: second)
		TS Time stamp. Display DT and TM together such as "YYYYMMDDHHMMSS."
	CM	Field of combined multiple data by component section separator character.
5	Maximum length (Max)	Maximum number of effective characters except escape characters in the relevant field.
6	Elecsys ® format effective (EV)	Indicates if the field is effective or not in record. Fields without X is ignored when received though they are defined by ASTM.
7	cobas ® format effective (CV)	R = Required field X = Effective field, if available
8	Comment (Comments)	Field description

Message header record

Elecsys® type format (upload, download)

```
H|\^&|||xxx|||||P|[CR]
```

(1) (2) (3) (6)

cobas® type format (upload, download)

```
H|\^&|||cobas-e411^1||||host|RSUPL^BATCH|P|1[CR]
```

(1) (2) (3) (4) (5) (6)(7)

Pos	No	Field	Type	Max	E V	C V	Comments	
							Elecsys® Type Format	cobas® Type Format
1	(1)	Record Type ID	ST	1	X	R	"H" fixed.	
2	(2)	Delimiter Definition	ST	4	X	R	Four characters such as field separator character, repeat separator character, component separator character, and escape character are defined. The first character is a field separator character and Record Type ID separator as well. These four characters are " ^&."	
3		Message Control ID						
4		Access Password						
5	(3)	Sender Name or ID	CM	36	X	X	Indicates sender name. (Can be omitted) Delete it when sending from cobas e 411 analyzer.	Setting is as follows. (Can be omitted) <Sender's device name>^<Communication program version> <Sender's device name> Type:TX Max: 30 Sending from cobas e 411 analyzer.: " cobas-e411 " fixed. Sending from HOST: any characters within alphanumeric and "-." <Communication program version> Type: NM Max: 5 "1" fixed.
6		Sender Street Address						
7		Reserved Field						
8		Sender Telephone Number						
9		Characteristics of Sender						
10	(4)	Receiver ID	ST	30		X		Receiver's name. (Can be omitted) Sending from cobas e 411 analyzer.: "host" fixed. Sending from HOST: any characters within alphanumeric and "-."
11	(5)	Comment or Special Instructions	CM	11		R		Setting is as follows: <Meaning of message>^<Cause> <Meaning of message> Type: ST Max: 5 "TSREQ": TS inquiry. "RSUPL": Transmitting results. "TSDWN": Test request. <Cause of message> Type: ST Max: 5 "REAL": communication in real time. "BATCH": communication based on request from cobas e 411 /HOST. "REPLY": replay to the request.
12	(6)	Processing ID	ST	1	X	R	"P" fixed.	
13	(7)	Version No.	NM	1		R		"1" fixed.
14		Date and Time of Message						

Message termination record

Elecsys® format (upload, download), **cobas®** format (upload, download)

L|1|N [CR]

(1)(2)(3)

Pos	No	Field	Type	Max	E V	C V	Comments	
							Elecsys® Type Format	cobas® Type Format
1	(1)	Record Type ID	ST	1	X	R	"L" fixed.	
2	(2)	Sequence Number	NM	6	X	R	Indicates sequence No. Normally it is "1"	
3	(3)	Termination Code	ST	1	X	R	Indicates the end of communication record. (Can be omitted). See Table E-2 Termination Code List for setting.	"N" : Normal end "E" : Receiving error, hardware error, application error

Table E-2 Termination code list (Elecsys® type format)

Message	Message Communication Status		Termination Code
Inquiry	Normal	With response data	"F"
		Without response data	"I"
	Abnormal	All data in record is not defined (inapplicable message error)	"Q"
		Receiving error Hardware error Application error	"E"
Response, upload, download	Normal		-(*)
	Abnormal	All data in record are not defined (inapplicable message error) Receiving error Hardware error Application error	"E"
Invalid Record	Abnormal	Receiving error Hardware error Application error	"E"

(*) The device does not transmit Termination Code when response or receiving message at download is normal.

Request information record

Elecsys® type (upload)

Q|1|^000663^32^@7^2^^SAMPLE^NORMAL||ALL|||||||O[CR]

(1) (2) (3) (4) (5)

cobas® type (upload, download)

Q|1|^000663^32^@7^2^^S1^SC||ALL|||||||O[CR]

(1) (2) (3) (4) (5)

Pos	No	Field	Type	Max	E V	C V	Comments	
							Elecsys® Type Format	cobas® Type Format
1	(1)	Record Type ID	ST	1	X	X	"Q" fixed.	
2	(2)	Sequence Number	NM	6	X	X	Indicates sequence No. Normally it is "1"	
3	(3)	Starting Range ID Number	CM	55 / 46	X	X	Indicates inquired sample. Setting is as follows: ^<SampleID>^<SequenceNo>^ <CarrierNo>^<PositionNo>^^ <SampleType>^<ContainerType> <SampleID> Type: ST Max: 22 indicates Sample No. (Sample ID.) <SequenceNo> Type: NM Max: 4 indicates e411 internal sequence No. <CarrierNo> Type: ST Max: 5 indicates carrier No. (Disk/Rack) <PositionNo> Type: NM Max: 2 indicates position No. in carrier. <SampleType> Type: ST Max: 7 indicates sample type. "SAMPLE": patients sample, static. <ContainerType> Type: ST Max: 7 indicates sample cup type. "NORMAL": test tube or sample cup. "REDUCED": sample cup, only.	Indicates inquired sample. Setting is as follows: ^^<SampleID>^<SequenceNo>^ <CarrierNo>^<PositionNo>^^ <SampleType>^<ContainerType> <SampleID> Type: ST Max: 22 (*1) indicates Sample No. (Sample ID.) <SequenceNo> Type: NM Max: 4 indicates e411 internal sequence No. <CarrierNo> Type: ST Max: 5 (*2) indicates carrier No. (Disk/Rack.) <PositionNo> Type: NM Max: 2 indicates position No. in carrier. <SampleType> Type: ST Max: 2 (*3) indicates sample type. "S1": blood serum. "S2": urine. "S5": others. <ContainerType> Type: ST Max: 7 (*4) indicates only sample cup type (SC). "SC": test tube or sample cup. "MC": reduced sample volume
4		Ending Range ID Number						
5	(4)	Universal Test ID	ST	3	X	X	"ALL" fixed.	
6		Nature of Request Time Limits						
7		Beginning Request Results Date and Time						
8		Ending Request Results Date and Time						
9		Requesting Physician Name						
10		Requesting Physician Telephone Number						
11		User Field No.1						
12		User Field No.2						

13	(5)	Request Information Status Codes	ST	1	X	X	Indicates the objective of the record. Setting is as follows: "O": Order query (to Host) "A": Cancel the last request (to Host)
----	-----	----------------------------------	----	---	---	---	---

Note (*1): When barcode read error occurs, the sample id will be generated as follows:

"@"<Sequence no.>.

Note (*2): When rack No. of rack version is unknown, "@" is attached in front of <CarrierNo> followed by a delivered value.

Note (*3): Sample Type "S1, S2, S5" must be changed to "S0" in the next software version because there is no rack type/sample type differentiation on the analyzer. The correct Sample Type will be send in the order record from the HOST.

Note (*4): MC indicates reduced volume not micro cup (like cobas 6000).

Patient information record

Elecsys® type (upload)

P 1	[CR]
-----	------

(1)(2)

Elecsys® type (download), cobas® format (upload, download)

P 1	[CR]
-----	------

(1)(2)

Pos	No	Field	Type	Max	EV	C V	Comments	
							Elecsys® Type Format	cobas® Type Format
1	(1)	Record Type ID	ST	1	X	R	"P" fixed.	
2	(2)	Sequence Number	NM	6	X	R	Indicates sequence No. Normally it is "1"	
3		Practice Assigned Patient ID						
4		Laboratory Assigned Patient ID						
5		Patient ID No. 3						
6		Patient Name						
7		Mother's Maiden Name						
8		Birthdates						
9		Patient Sex						
10		Patient Race						
11		Patient Address						
12		Reserved Field						
13		Patient Phone No						
14		Attending Physician ID						
15		Special Field 1						

Test order record

Elecsys® type (upload)

O|1|000663|36^0044^2^^SAMPLE^NORMAL|ALL|R|20050705093416||||X|||||||O [CR]
 (1)(2) (3) (4) (5) (6) (7) (8) (11)

Elecsys® type (download)

O|1|000663|36^0044^2^^SAMPLE^NORMAL|^^^10^2|R|20050705093416||||N|||||||Q [CR]
 (1)(2) (3) (4) (5) (6) (7) (8) (11)

cobas® type (upload)

O|1|000663|6^44^2^^S1^SC|^^^10^2|R||20050705093416||||N|||1|||||20050705095504||F[CR]
 (1)(2) (3) (4) (5) (6) (7) (8) (9) (10) (11)

cobas® type (download)

O|1|000663|6^44^2^^S1^SC|^^^10^2|R||20050705093416||A|||1|||||O[CR]
 (1)(2) (3) (4) (5) (6) (7) (8) (9) (11)

Pos	No	Field	Type	Max	E V	C V	Comments	
							Elecsys® Type Format	cobas® Type Format
1	(1)	Record Type ID	ST	1	X	R	"O" fixed.	
2	(2)	Sequence Number	NM	6	X	R	Indicates sequence No. Normally it is "1"	
3	(3)	Specimen ID	ST	22	X	R	Indicates sample No. (Sample ID.) For control sample, set name of control.	Indicate sample No. (Sample ID.) When barcode read error occurs, the sample id will be generated as follows: "@"<Sequence no.> For control sample, the "control name" or "control name^lot number" is sent to the host, according to the settings at the host communication setting screen.
4	(4)	Instrument Specimen ID	CM	30 / 20	X	R	Indicates ordered sample. Setting is as follows: <SequenceNo>^<CarrierNo>^<PositionNo>^^<SampleType>^<ContainerType> <SequenceNo> Type: NM Max: 4 indicates cobas e 411 analyzer. internal sequence No. <CarrierNo> Type: ST Max: 5 indicates carrier No. (Disk/Rack.) <PositionNo> Type: NM Max: 2 indicates position No. in carrier. <SampleType> Type: ST Max: 7 indicates sample type. "SAMPLE": patient sample. "CONTROL": control sample. <ContainerType> Type: ST Max: 7 indicates type of sample cup. "NORMAL": test tube or sample cup. "REDUCED": sample cup, only.	Indicates ordered sample. Setting is as follows: <SequenceNo>^<CarrierNo>^<PositionNo>^^<SampleType>^<ContainerType> <SequenceNo> Type: NM Max: 4 indicates cobas e 411 analyzer. internal sequence No. <CarrierNo> Type: ST Max: 5 indicates carrier No. (Disk/Rack.) <PositionNo> Type: NM Max: 2 indicates position No. in carrier. <SampleType> Type: ST Max: 2 (*2) indicates sample type "S1": blood serum. "S2": urine. "S5": others. "QC": control sample. <ContainerType> Type: ST Max: 7 (*1) indicates type of sample cup. "SC": test tube or sample cup. "MC": reduced sample volume.

5	(5)	Universal Test ID	CM	8	X	X	Indicates order. Repeats up to 60 orders. Only up to 18 for test selection information are available. Setting is as follows: ^^^<ApplicationCode>^<Dilution>\... Repeat delimiter \ for multiple test selection. For upload, "ALL" fixed. <ApplicationCode> Type: NM Max: 3 (*4) indicates e411 Application No. <Dilution> Type: NM Max: 1 indicates automatic dilution factor. (*5) "0": ratio=1 (not diluted). "1": ratio=2 "2": ratio=5 "3": ratio=10 "5": ratio=20 "7": ratio=50 "9":ratio=100	Indicates order. Repeats up to 60 orders. Only up to 18 for test selection information are available. Setting is as follows: ^^^<ApplicationCode>^<Dilution>\... Repeat delimiter \ for multiple test selection. <ApplicationCode> Type: NM Max: 3 (*4) indicates e411 host test No. The host test no. can be a test code or a calculated test no (only for upload). <Dilution> Type: NM Max: 1 (*5) indicates automatic dilution ratio. (also for no dilution attach "^" after Host Test No. like "^^^10^") None or "1": ratio=1 (not diluted) "2": ratio=2 "5": ratio=5 "10": ratio=10 "20": ratio=20 "50": ratio=50 "100": ratio=100
6	(6)	Priority	ST	1	X	R	Indicates priority of patient samples. "R": routine, control sample. "S": stat sample	Indicates priority of patient samples. Not used for control samples. "R": routine sample. "S": stat sample.
7		Requested/Ordered Date and Time						
8	(7)	Specimen Collection Date and Time	TS	14	X	X	Indicates reception date and time of request. Setting is as follows. Optional YYYYMMDDHHMMSS	
9		Collection End Time						
10		Collection Volume						
11		Collector ID						
12	(8)	Action Code	ST	2 / 1	X	R	Indicates current status of sample (*3) "X": measured (upload) "N": new sample order (download) "X\Q": control sample (upload) "Q": control sample. (download) "A": additional test order (upload) Indicates type of information to be reported (*3) "N": communication of patient sample result from analyzer. (upload) "Q": communication of control sample result from analyzer (upload) "A": test order from HOST. (download)	
13		Danger Code						
14		Relevant Clinical Information						
15		Date/Time Specimen Received						
16	(9)	Specimen Descriptor	NM	1		R	Indicates sample type (*2) "1": blood serum. "2": urine. "5": others.	
17		Ordering Physician						
18		Physician's Telephone Number						
19		User Field No.1						
20		Users Field No.2						
21		Laboratory Field No.1						
22		Laboratory Field No.2						

23	(10)	Date/Time Results Reported or Last Modified	TS	14		X		Indicates date when all test results are collected. Setting from HOST is not applicable. Setting is as follows: YYYYMMDDHHMMSS
24		Instrument Charge to Computer System						
25		Instrument Section ID						
26	(11)	Report Types	ST	1	X	R	Indicates report type. "Q": response to inquiry. (download) "Z": no response request to inquiry. (download) "O": from e411 to HOST. (upload)	Indicates type of communication. "O": test order. (download) "F": communication of result. (upload)
27		Reserved Field						
28		Location or Ward of Specimen Collection						
29		Nosocomial Infection Flag						
30		Specimen Service						
31		Specimen Institution						

Note (*1): MC indicates reduced volume not micro cup (like cobas 6000).

Note (*2): Except for quality controls, the Specimen Descriptor of Instrument Specimen ID for **cobas®** type is prior to Sample Type when they are mismatched.

Note (*3): Action Code of already ordered sample for **Elecsys®** type is "N": When new order of sample is received, delete existing test information to order the receiving tests. Same for **cobas®** type as its Action Code is "A" fixed (download).

Note (*4): Application Code in Universal Test ID

Elecsys® type

The last digit of application no is called "generation" and the host should handle application no with taking "generation" into account.

If the host downloads an order of TSH with application no=10 and only a different generation of TSH (application no=11) is on board then **cobas e 411** analyzer uses TSH (application no=11) for determination and sends back the result with application no=11.

cobas® type

One host test code can be assigned to two or more test generations (e.g., TSH 0(10), TSH 1(11)). In this case, the latest generation on board will be the highest priority for reagent pack selection.

Note (*5): See Table E-3 for automatic dilution factor/ratio for <Dilution> of Universal Test ID.

Table E-3 Automatic dilution factor/ratio list

Automatic dilution Factor (Elecsys)	Automatic dilution ratio (Cobas)	Pipetting volumes			
		1 st dilution		2 nd dilution	
		Sample [volume]	Diluent [volume]	Diluted sample [volume]	Diluent [volume]
0	1	No sample dilution			
1	2	50 µl	50 µl	-	-
2	5	40 µl	160 µl	-	-
3	10	20 µl	180 µl	-	-
5	20	20 µl	180 µl	100 µl	100 µl
7	50	20 µl	180 µl	40 µl	160 µl
9	100	20 µl	180 µl	20 µl	180 µl

Result record

Elecsys® type (upload)

R		1		^^^10^^0		0.310		uIU/ml		0.270^4.20		N		F			20050619094203		20050619101521	[CR]
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)

cobas® type (upload)

R		1		^^^10//not		0.310		uIU/ml		N		F			admin			E1	[CR]
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)

Pos	No	Field	Type	Max	E V	C V	Comments	
							Elecsys® Type Format	cobas® Type Format
1	(1)	Record Type ID	ST	1	X	X	"R" fixed.	
2	(2)	Sequence Number	NM	6	X	X	Indicates sequence No.	
3	(3)	Universal Test ID	CM	10 / 20	X	X	Indicates order. ^^^<ApplicationCode>^<Dilution>^ <pre-dilution> <ApplicationCode> Type: NM Max: 3 indicates e411 Application No. <Dilution> Type: NM Max: 1 indicates automatic dilution factor when ordering. < pre-dilution > Type: ST Max: 1 "0": without pre-dilution. "1": with pre-dilution.	Indicates order. ^^^<ApplicationCode>/<Dilution>/ <pre-dilution> <ApplicationCode> Type: NM Max: 3 indicates e411 host test no. The host test no. can be a test code or a calculated test no.. Calculated tests can be defined on the "Calc. Test" screen. A total of 5 calculated tests can be set. When all tests requiring calculation are ordered and successfully measured, the analyzer automatically transmits the calculated tests. However, the host may not send a calculation test order. <Dilution> Type: ST Max: 1 indicates automatic dilution factor when ordering: 1, 2, 5, 10, 20, 50, 100 < pre-dilution > Type: ST Max: 11 "not": without pre-dilution. "pre-diluted": with pre-dilution.

4	(4)	Data or Measurement Value	CM	10	X	X	Indicates measured value. Quantitative test format: <measurement value> Qualitative test format: <measurement value> ^ <cut off index> <measurement value> Type: NM Max: 7 for quantitative tests: seven numeric including symbol and decimal places. ">", "<" is attached to the top of the measured value when measuring range error occurred. Seven spaces (0x20) are communicated if no result. For qualitative tests: "1": Positive "0": Border line "-1": Negative <cut off index> Type: NM Max: 7 seven numeric including symbol and decimal places. Seven spaces (0x20) are communicated if no result.	Indicates measured value. Quantitative test format: <measurement value> > ^ <message value> Qualitative test format: <qualitative value> ^ <cut off index> <measurement value> Type: NM Max: 7 measured value or 7 spaces if no result. ">", "<" is attached before the measured value when measuring range error occurred. 7 digit including symbol and decimal places. <qualitative value> Type: NM Max: 2 Result message "1": Positive (reac.) "0": Border line (border) "-1": Negative (n-reac.) <message value> Type : NM Max : 2 Result Message code (0 ~ 31) <cut off index> Type: NM Max: 7 Measured value or 7 spaces if no result. ">", "<" is attached before the measured value when measuring range error occurred. 7 digit including symbol and decimal places.
5	(5)	Units	ST	6	X	X	Indicates unit name of measurement results.	
6	(6)	Reference Ranges	CM		X		Indicates normal range. Indicates QC range when control sample. <Low> ^ <High> <Low> Type: NM Max: 7 indicates minimum of the normal range. <High> Type: NM Max: 7 indicates maximum of the normal range.	
7	(7)	Result Abnormal Flags	ST	2	X	X	Indicates normal/abnormal of measurement results. "L": less than normal range. "H": more than normal range. "<": less than measured range. ">": more than measured range. "N": Normal "A": Abnormal	Indicates normal/abnormal of measurement results. "L": less than normal range. "H": more than normal range. "LL": less than measured value. "HH": more than measured value. "N": Normal. "A": Abnormal.
8		Nature of Abnormality Testing						
9	(8)	Result Status	ST	1	X	X	Indicates the number of the test conducted for the analytical data. "F": last result. "X": results cannot be done, "R": the result communicated. "V": released result by user "Y": blocked by system. "+": blocked by user.	Indicates the number of the test conducted for the analytical data. "F": initial result. "C": rerun result.
10		Date of Change in Instrument Normative values Units						
11	(9)	Operator Identification	ST	6		X		Indicates operator ID who conducted measurement. HOST can't set this field.
12	(10)	Data/Time Test Started	TS	14	X		Indicates time and date when starting measurement. Setting is as follows: YYYYMMDDHHMMSS	

13	(11)	Date/Time Test Completed	TS	14	X	Indicates time and date when completing measurement. Setting is as follows. YYYYMMDDHHMMSS	
14		Instrument Identification					"E1"

Note 1 :

cobas® type format transmits calculated tests. Elecsys® type format does not transmit calculated tests.

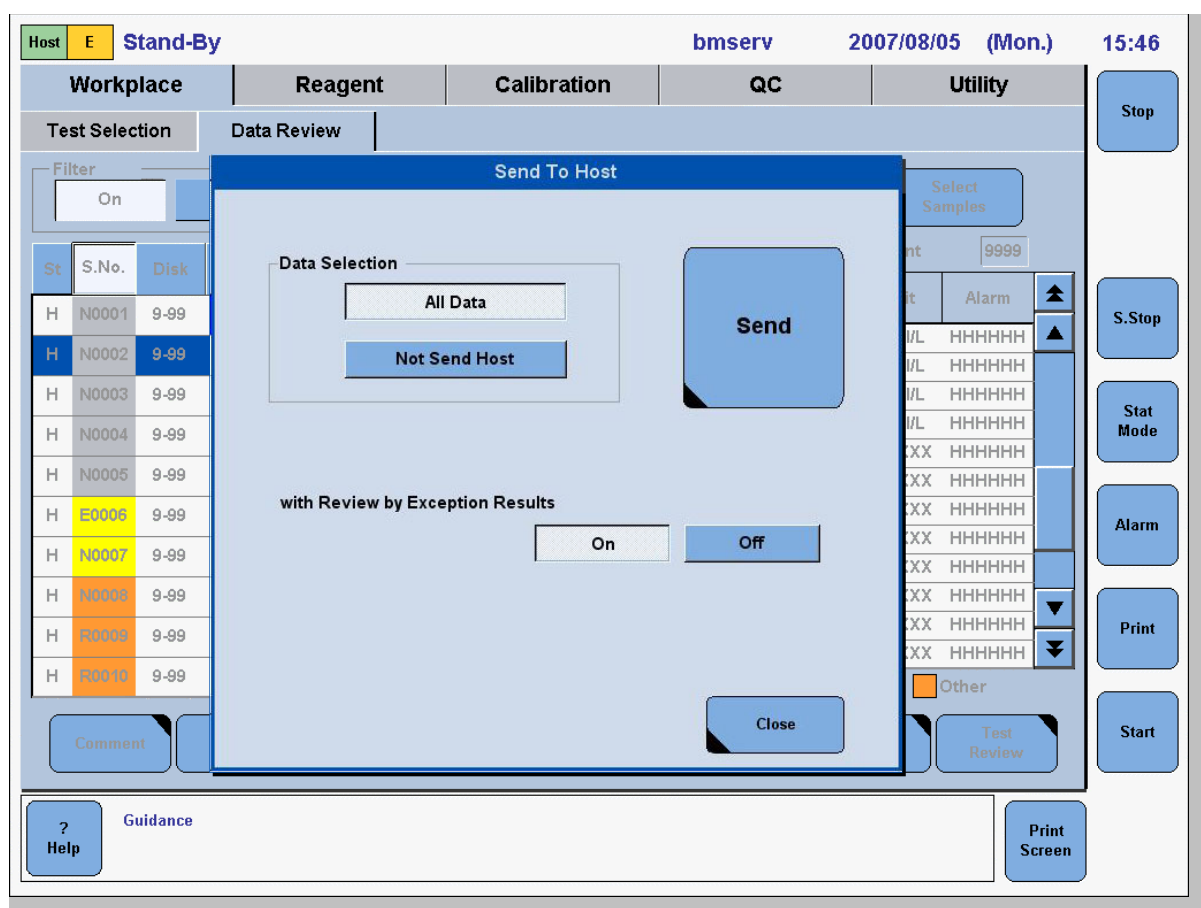
Note 2 :

Elecsys® type format has no "Review by exception" function.

Cobas® type format does not transmit tests specified by exception alarm, defined in the "Review by exception" screen.

Communication of batch results with review by exception alarm flags is based on option "with Review by Exception Results" on [Send to Host] window shown below.

On: tests specified by Exception Alarm are communicated.



Off: tests specified by Exception Alarm are not communicated.

When "Not Send Host" is selected on the window "Send to Host", only samples which have not been communicated to the host, will be transmitted.

Comment record



The comment record, that follows the order record for patient demographic data is not used.

Result flag [C-RES] "(Comment Record, that follows the Result Record)"

Elecsys® type (upload)

C	1	I	I	50	^Below measuring range	I	[CR]
---	---	---	---	----	------------------------	---	------

(1)(2)(3) (4) (5)

cobas® type (upload)

C	1	I	I	50	I	[CR]
---	---	---	---	----	---	------

(1)(2)(3)(4)(5)

Pos	No	Field	Type	Max	E V	C V	Comments	
							Elecsys® Type Format	cobas® Type Format
1	(1)	Record Type ID	ST	1	X	X	"C" fixed.	
2	(2)	Sequence Number	NM	6	X	X	Indicates sequence No. Normally it is "1"	
3	(3)	Comment Source	ST	1	X	X	"I" fixed.	
4	(4)	Comment Text	NM / CM	53 / 3	X	X	Data alarm No, and message for the measured value is attached. <Alarm Flag> ^ <Alarm Messages> < Alarm Flag > Type: NM Max: 2 indicates alarm No. <AlarmMessages>Type: ST Max: 50 indicates alarm message.	Data alarm no. for the measured value is attached. (*1)
5	(5)	Comment Type	ST	1	X	X	"I" fixed.	



Skip this record when no data alarm for message value was produced.

Note (*1): See Table E-5 for data alarm no. and alarm message. Alarm message does not correspond to UTF-8.

Table E-5 Data alarm list

Flag	Data Alarm	Screen/ Printer	Elecsys® Type Host No	cobas® Type Host No
1	Power Fail/Power Off Cancel	Cancel	1	-
2	E.STOP Cancel	Cancel	2	-
3	STOP Cancel	Cancel	3	-
4	P.STOP/A.STOP Cancel	Cancel	4	-
5	S.STOP Cancel	Cancel	5	-
6	Recovery Cancel	Cancel	6	-
7	Sample Short	Samp.S	7	3
8	Assay Reagent Short	Reag.S	8	4
9	Diluent Short	Reag.S	9	4
10	Pretreatment Reagent Short	Reag.S	10	4
11	<Not available>	-	-	-
12	Abnormal Reagent Disk Temperature	Reag.T	12	74
13	Abnormal Incubator Temperature	Inc.T	13	75
14	Abnormal Measuring Cell Temperature	Cell.T	14	77
15	Abnormal System Reagent Temperature	SysR.T	15	76
16	System Reagent Short	SysR.S	16	62
17	ADC abnormal	ADC.E	17	1
18	<Not available>	-	-	-
19	<Not available>	-	-	-
20	<Not available>	-	-	-
21	<Not available>	-	-	-
22	<Not available>	-	-	-
23	<Not available>	-	-	-
24	Calculation Error	Calc.?	24	39
25	No Calibration Data	Cal.E	25	43
26	Previous Calibration Data	Cal.E	26	43
27	<Not available>	-	-	-
28	<Not available>	-	-	-
29	<Not available>	-	-	-
30	<Not available>	-	-	-
31	Assay Reagent Hovering	Reag.H	31	69
32	Diluent Hovering	Reag.H	32	69
33	Pretreatment Hovering	Reag.H	33	69
34	<Not available>	-	-	-
35	Assay Reagent Film Detected	Reag.F	35	70
36	Diluent Film Detected	Reag.F	36	70
37	Pretreatment Film Detected	Reag.F	37	70
38	System Reagent Film Detected	Reag.F	38	70
39	<Not available>	-	-	-
40	AB Level Range Over	>AB	40	63
41	AB Level Check Error	AB.E	41	64
42	Current Range Over	>Curr	42	65
43	Current Range Check Error	Curr.E	43	66
44	System Reagent Temperature Unstable	SysR.U	44	120
45	Sample Clot Detected	Samp.C	45	72
46	Potential microparticle carryover	CarOvr	46	71
47	Sample ID Error Cancel	Cancel	47	121
48	Below normal(expected)range	L	48	41
49	Above normal(expected)range	H	49	40
50	Below measuring range	<Test	50	27
51	Above measuring range	>Test	51	26
52	Expired RackPack	ReagEx	52	101
53	No Sample	Samp.S	53	3
54	Sample LLD Inexecution	SLLD.E	54	86
55	Sample LLD Noise	SLLD.N	55	87
56	Current Range Over(Operation)	>Curr	56	99
57	Instrument Factor A	FacA	57	122
58	Signal level below limit	<SigL	58	100
59	Calc Test Error	ClcT.E	-	37
-	Overflow (*)	Over.E	-	38

(*) Not in use

Communication trace

F

- 1 *Test selection information in real time* F-2
- 2 *Real time test results*..... F-6
- 3 *Batch test selection information* F-9

Communication trace

Example of communication trace is shown below.

The following trace is shown without Start ([STX][FN]), End ([CR][EXT][CS1][CS2][CR][LF]), and response from each receiver ([ENQ][ACK][EOT]).

Test selection information in real time



HOST is required to send back SequenceNo, CarrierNo, and Position to cobas e 411 analyzer without changing them.

(1) Disk Type

a) [Elecsys® type]

cobas e 411 analyzer sends test selection information such as sample ID=000004, sequence No.=40, Disk No=0, and Position=5.

```
H|\^&|||||P||[CR]
Q|1|^000004^40^0^5^^SAMPLE^NORMAL||ALL|||||O[CR]
L|1|[CR]
```

Example1 : HOST replies test selection information of sample ID=000004.

```
H|\^&|||||P||[CR]
P|1 [CR]
O|1|000004|40^0^5^^SAMPLE^NORMAL|^10^30^2^40^|R||||N
|||||Q
L|1|[CR]
```

Example 2 : HOST replies without order information of sample ID=000004.

```
H|\^&|||||P||[CR]
P|1 [CR]
O|1|000004|40^0^5^^SAMPLE^NORMAL||R||||N|||||Z[CR]
L|1|[CR]
```

cobas e 411 analyzer sends a cancel to HOST when HOST does not replies within 15 seconds.

```
H|\^&|||||P||[CR]
Q|1|^000004^40^0^5^^SAMPLE^NORMAL||ALL|||||A[CR]
L|1|[CR]
```

b) [**cobas**® type]

cobas e 411 analyzer sends test selection information such as sample ID=000004, sequence No.=40, Disk No=0, and Position=5.

```
H|\^&|||cobas-e411^1||||host|TSREQ^REAL|P|1[CR]
Q|1|^000004^40^0^5^^S1^SC||ALL|||||O[CR]
L|1|N[CR]
```

Example 1 : HOST replies test selection information of sample ID=000004.

```
H|\^&|||host^1||||cobas-e411|TSDWN^REPLY|P|1[CR]
P|1 [CR]
O|1|000004|40^0^5^^S1^SC|^10^\^^30^2\^^40^|R|||||A
|||1|||||||O[CR]
L|1|N[CR]
```

Example 2 : HOST replies without order information of sample ID=000004 or the generated sample id=@<sequence no.>.

```
H|\^&|||host^1||||cobas-e411|TSDWN^REPLY|P|1[CR]
P|1 [CR]
O|1|000004|40^0^5^^S1^SC||R|||||A|||1|||||||O[CR]
L|1|N[CR]
```

cobas e 411 analyzer sends a cancel to HOST when HOST does not replies within 15 seconds.

```
H|\^&|||cobas-e411^1||||host|TSREQ^REAL|P|1[CR]
Q|1|^000004^40^0^5^^S1^SC||ALL|||||A[CR]
L|1|N[CR]
```

(2) Rack Type

a) [Elecsys® type]

cobas e 411 analyzer sends inquiry for sample ID=000002, sequence No.=3 Rack No.=@95 and Position=2.

```
H|^&|||||||P||[CR]
Q|1|^000002^3^@95^2^^SAMPLE^NORMAL||ALL|||||||O[CR]
L|1|[CR]
```

HOST replies test selection information of sample ID=000002.

```
H|^&|||||||P||[CR]
P|1|[CR]
O|1|000002|3^@95^2^^SAMPLE^NORMAL|^^^10^|R|||||N|||||||Q[CR]
L|1|[CR]
```

cobas e 411 analyzer sends inquiry for sequence No.=3 Rack No.=0007, Position=2, and no sample ID.

```
H|^&|||||||P||[CR]
Q|1|^3^0007^2^^SAMPLE^NORMAL||ALL|||||||O[CR]
L|1|[CR]
```

Note: Key information for a non-barcoded sample is Rack No and position. If barcode read error takes place for a barcoded sample, the key information is also Rack No and position. In such cases, **cobas e 411** analyzer inquires after reading Rack No.

b) [cobas® type]

cobas e 411 analyzer sends inquiry for sample ID=000002, sequence No.=3 Rack No.=@95 and Position=2.

```
H|^&|||cobas-e411^1||||host|TSREQ^REAL|P|1|[CR]
Q|1|^000002^3^@95^2^^S1^SC||ALL|||||||O[CR]
L|1|N[CR]
```

HOST replies test selection information of sample ID=000002.

```
H|^&|||host^1||||cobas-e411|TSDWN^REPLY|P|1|[CR]
P|1|[CR]
O|1|000002|3^@95^2^^S1^SC|^^^10^|R|||||A||||1|||||||O[CR]
L|1|N[CR]
```

cobas e 411 analyzer sends inquiry for sequence No.=3 Rack No.=0007, Position=2 and a generated sample id=@3.

<pre>H \^& cobas-e411^1 host TSREQ^REAL P 1[CR] Q 1 ^@3^3^0007^2^^S1^SC ALL O[CR] L 1 N[CR]</pre>
--

Note: When it is rack type without sample ID, incl. barcode read error, **cobas e 411** analyzer inquires after reading Rack No.



If a barcode read error occurs and the rack no. has been generated, the sample can not be identified by the host.

When the rack type communication for a rack does not complete within approximately 40 seconds, the cobas e 411 analyzer sends a cancel to HOST.

Real time test results

(1) When the result value is within normal range.

a) [Elecsys® type]

cobas e 411 analyzer sends test result of sample ID=000004, sequence No.=40, Disk No.=0 and Position=5.

```
H|\^&|||||P||[CR]
P|1||||| [CR]
O|1|000004|40^0^5^^SAMPLE^NORMAL|ALL|R|20051220095504||||X
|||||O[CR]
R|1|^^^10^^0|1.25|uIU/ml|0.270^4.20|N||F||20051220095534|
20051220101604|[CR]
R|2|^^^30^2^1|1.52|ng/dl|1.01^1.79|N||F||20051220103034|
20051220105004|[CR]
R|3|^^^40^^0|1.17|uIU/ml|0.846^2.02|N||F||20051220110034|
20051220112004|[CR]
L|1|[CR]
```

b) [cobas® type]

cobas e 411 analyzer sends test result of sample ID=000004, sequence No.=40, Disk No.=0 and Position=5.

```
H|\^&|||cobas-e411^1|||||host|RSUPL^REAL|P|1|[CR]
P|1 [CR]
O|1|000004|40^0^5^^S1^SC|^^^10^^^^30^2\^^^40^|R||||N
|||1|||||20051220095504|||F[CR]
R|1|^^^10//not|1.25^|uIU/ml||N||F||admin|||E1[CR]
R|2|^^^30/2/pre-diluted|0.091^|ng/dl||N||F||admin|||E1[CR]
R|3|^^^40//not|1.17^|ng/ml||N||F||admin|||E1[CR]
L|1|N[CR]
```

(2) When the result value is less than normal range.

a) [Elecsys® type]

cobas e 411 analyzer sends a test result of sample ID=000002, sequence No.=3, Rack No.=0007, and Position=2.

```
H|\^&|||||||P||[CR]
P|1||||||| [CR]
O|1|000002|3^0007^2^^SAMPLE^NORMAL|ALL|R|20051220104418||||X
|||||||O[CR]
R|1|^^^10^^0|0.163|uIU/ml|0.270^4.20|L||F|||20051220103034|
20051220105004|[CR]
C|1||48^Below normal(expected) range|[CR]
L|1|[CR]
```

b) [cobas® type]

cobas e 411 analyzer sends a test result of sample ID=000002, sequence No.=3, Rack No.=0007, and Position=2.

```
H|\^&|||cobas-e411^1|||||host|RSUPL^REAL|P|1[CR]
P|1 [CR]
O|1|000002|3^0007^2^^S1^SC|^^^10^|R|||||N||||1|||||
20051220104418|||F[CR]
R|1|^^^10//not|0.163|uIU/ml||L||F||admin|||E1[CR]
C|1||48|[CR]
L|1|N[CR]
```

(3) When the result value is a qualitative test.

a) [Elecsys® type]

cobas e 411 analyzer sends a test result of sample ID=000010, sequence No.=442, Rack No.=0005, and Position=1.

```
H|^&|||||P||[CR]
P|1||||| [CR]
O|1|000010|442^0005^1^^SAMPLE^NORMAL|ALL|R|20051220104418||||X
|||||O[CR]
R|1|^^^400^^0|-1^0.303|COI|^|N||F||20051220110334|
20051220112404|[CR]
L|1|[CR]
```

b) [cobas® type]

cobas e 411 analyzer sends a test result of sample ID=000010, sequence No.=442, Rack No.=0005, and Position=1.

```
H|^&|||cobas-e411^1||||host|RSUPL^REAL|P|1[CR]
P|1 [CR]
O|1|000010|442^0005^1^^S1^SC|^^^400^|R||||N||||1|||||
20051220104418||F[CR]
R|1|^^^400//not|-1^0.303|COI||N||F||admin|||E1[CR]
L|1|N[CR]
```

(4) When it is a control sample.

a) [Elecsys® type]

```
H|^&|||||P||[CR]
P|1||||| [CR]
O|1|PC U2|96^0019^1^^CONTROL^NORMAL|ALL|R|20051220104418||||X\Q
|||||O[CR]
R|1|^^^10^^0|1.45|uIU/ml|1.37^1.97|N||F||20051220110334|
20051220112404|[CR]
L|1|[CR]
```

b) [cobas® type]

```
H|^&|||cobas-e411^1||||host|RSUPL^REAL|P|1[CR]
P|1 [CR]
O|1|PC U2|96^0019^1^^QC^SC|^^^400^|||||Q||||1
|||||20051220104418||F[CR]
R|1|^^^400//not|1.26^|uIU/ml||L||F||admin|||E1[CR]
L|1|N[CR]
```

Batch test selection information

Download test selection information of sample ID=000051 from HOST. [**cobas®** type]

```
H|\^&|||host^1|||cobas-e411|TSDWN^BATCH|P|1[CR]
P|1 [CR]
O|1|000051|^^^S1^SC|^^10^/^^30^2/^^40^|R||||A||||1
|||||||O[CR]
L|1|N[CR]
```



Download of test selection information is new order only. Sequence No. is automatically numbered by the cobas e 411 analyzer.

The cobas e 411 analyzer can store 2000 tests, e.g. for 1000 samples with 2 tests. The older tests will be overwritten with new order if 2000 tests are already stored.

Communication error

G

1 *Example*G-2

Communication error

Example

Example of communication error is shown below.

Table G-1 Communication error list

Content	Alarm Code
A try occurred at message transmission	44-01-01
Communication abort at mess. transmission	44-01-02
Communication abort at receiving message	44-01-03
Message retransmission was unsuccessful (others)	44-01-04
Message retransmission was unsuccessful (retry failed)	44-01-05
Timeout occurred at message transmission	44-01-06
Timeout occurred while receiving message	44-01-07
Communication format was incorrect	44-01-08
Update of a database was not allowed	44-01-09
A hardware error occurred	44-01-10
A software error occurred	44-01-11
Upload is defined but host com. is OFF	44-01-12

Alarm messages are recorded in host trace log.

```

*****
Roche Diagnostics      Immunoanalyzer cobas e 411      S/N C703-37
*****
System Communication Trace  Operator-ID: Admin      2005/05/23, 12:35
-----

YY/MM/DD HH:MM:SS.msec
-----
06/02/20 13:48:32.187 S 001 <ENQ>
06/02/20 13:48:47.335 I 44-01-06 Timeout occurred at message transmission
06/02/20 13:48:47.277 I 44-01-12 Upload is defined but host com. is OFF
06/02/20 13:48:47.296 S 001 <EOT>
06/02/20 13:53:27.187 S 001 <ENQ>
06/02/20 13:53:27.296 R 001 <ACK>
06/02/20 13:53:27.377 S 041 <STX>1H|^~/&|||||||A.2|' '<CR><STX>38<CR><LF>

```

S : Send R : Recv E : Error

Appendix

H

1	<i>Control characters</i>	<i>H-2</i>
2	<i>Printable characters</i>	<i>H-3</i>

Control characters

Binary	Dec	Hex	Abbreviation	Description
0000 0000	0	00	NUL	Null character
0000 0001	1	01	SOH	Start of Header
0000 0010	2	02	STX	Start of Text
0000 0011	3	03	ETX	End of Text
0000 0100	4	04	EOT	End of Transmission
0000 0101	5	05	ENQ	Enquiry
0000 0110	6	06	ACK	Acknowledgment
0000 0111	7	07	BEL	Bell
0000 1000	8	08	BS	Backspace
0000 1001	9	09	HT	Horizontal Tab
0000 1010	10	0A	LF	Line feed
0000 1011	11	0B	VT	Vertical Tab
0000 1100	12	0C	FF	Form feed
0000 1101	13	0D	CR	Carriage return
0000 1110	14	0E	SO	Shift Out
0000 1111	15	0F	SI	Shift In
0001 0000	16	10	DLE	Data Link Escape
0001 0001	17	11	DC1	Device Control 1
0001 0010	18	12	DC2	Device Control 2
0001 0011	19	13	DC3	Device Control 3
0001 0100	20	14	DC4	Device Control 4
0001 0101	21	15	NAK	Negative Acknowledgement
0001 0110	22	16	SYN	Synchronous Idle
0001 0111	23	17	ETB	End of Trans. Block
0001 1000	24	18	CAN	Cancel
0001 1001	25	19	EM	End of Medium
0001 1010	26	1A	SUB	Substitute
0001 1011	27	1B	ESC	Escape
0001 1100	28	1C	FS	File Separator
0001 1101	29	1D	GS	Group Separator
0001 1110	30	1E	RS	Record Separator
0001 1111	31	1F	US	Unit Separator

0111 1111	127	7F	DEL	Delete
-----------	-----	----	-----	--------

Printable characters

Binary	Dec	Hex	Char	Binary	Dec	Hex	Char	Binary	Dec	Hex	Char
0010 0000	32	20	blank	0100 0000	64	40	@	0110 0000	96	60	`
0010 0001	33	21	!	0100 0001	65	41	A	0110 0001	97	61	a
0010 0010	34	22	"	0100 0010	66	42	B	0110 0010	98	62	b
0010 0011	35	23	#	0100 0011	67	43	C	0110 0011	99	63	c
0010 0100	36	24	\$	0100 0100	68	44	D	0110 0100	100	64	d
0010 0101	37	25	%	0100 0101	69	45	E	0110 0101	101	65	e
0010 0110	38	26	&	0100 0110	70	46	F	0110 0110	102	66	f
0010 0111	39	27	'	0100 0111	71	47	G	0110 0111	103	67	g
0010 1000	40	28	(	0100 1000	72	48	H	0110 1000	104	68	h
0010 1001	41	29	)	0100 1001	73	49	I	0110 1001	105	69	i
0010 1010	42	2A	*	0100 1010	74	4A	J	0110 1010	106	6A	j
0010 1011	43	2B	+	0100 1011	75	4B	K	0110 1011	107	6B	k
0010 1100	44	2C	,	0100 1100	76	4C	L	0110 1100	108	6C	l
0010 1101	45	2D	-	0100 1101	77	4D	M	0110 1101	109	6D	m
0010 1110	46	2E	.	0100 1110	78	4E	N	0110 1110	110	6E	n
0010 1111	47	2F	/	0100 1111	79	4F	O	0110 1111	111	6F	o
0011 0000	48	30	0	0101 0000	80	50	P	0111 0000	112	70	p
0011 0001	49	31	1	0101 0001	81	51	Q	0111 0001	113	71	q
0011 0010	50	32	2	0101 0010	82	52	R	0111 0010	114	72	r
0011 0011	51	33	3	0101 0011	83	53	S	0111 0011	115	73	s
0011 0100	52	34	4	0101 0100	84	54	T	0111 0100	116	74	t
0011 0101	53	35	5	0101 0101	85	55	U	0111 0101	117	75	u
0011 0110	54	36	6	0101 0110	86	56	V	0111 0110	118	76	v
0011 0111	55	37	7	0101 0111	87	57	W	0111 0111	119	77	w
0011 1000	56	38	8	0101 1000	88	58	X	0111 1000	120	78	x
0011 1001	57	39	9	0101 1001	89	59	Y	0111 1001	121	79	y
0011 1010	58	3A	:	0101 1010	90	5A	Z	0111 1010	122	7A	z
0011 1011	59	3B	;	0101 1011	91	5B	[	0111 1011	123	7B	{
0011 1100	60	3C	<	0101 1100	92	5C	\	0111 1100	124	7C	
0011 1101	61	3D	=	0101 1101	93	5D	]	0111 1101	125	7D	}
0011 1110	62	3E	>	0101 1110	94	5E	^	0111 1110	126	7E	~
0011 1111	63	3F	?	0101 1111	95	5F	_				

